

CHAPTER VI

PARTIAL DIFFERENTIATION

Consider a right circular cylinder with radius r and height h . Then the surface area of the cylinder is $S = 2\pi r^2 + 2\pi rh = f(r, h)$.

Partial derivative of S with respect to h (when r is held constant and h is allowed to vary) is $\frac{\partial S}{\partial h} = 2\pi r$. Similarly, the partial derivative of S with respect to r (when h is held constant and

r is allowed to vary) is $\frac{\partial S}{\partial r} = 4\pi r + 2\pi h$.

Examples

1. Given $f(x, y) = x^3y + x^2y^2 + 4x$. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution: $\frac{\partial f}{\partial x} = 3x^2y + 2xy^2 + 4$ and $\frac{\partial f}{\partial y} = x^3 + 2x^2y$.

2. Given $f(x, y, z) = x^2 + 2xy^2 + yz^3$. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution: $\frac{\partial f}{\partial x} = 2x + 2y^2$, $\frac{\partial f}{\partial y} = 4xy + z^3$ and $\frac{\partial f}{\partial z} = 3yz^2$.

3. Given $f(x, y) = \log(x + y)$. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution: $\frac{\partial f}{\partial x} = \frac{1}{x + y} \frac{\partial}{\partial x}(x + y) = \frac{1}{x + y}$. Similarly, $\frac{\partial f}{\partial y} = \frac{1}{x + y}$.

4. Given $f(x, y) = x \sin xy$. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution: $\frac{\partial f}{\partial x} = x \frac{\partial}{\partial x}(\sin xy) + \sin xy \frac{\partial}{\partial x}(x) = x \cos xy \frac{\partial}{\partial x}(xy) + \sin xy = xy \cos xy + \sin xy$.

Similarly, $\frac{\partial f}{\partial y} = x \frac{\partial}{\partial y}(\sin xy) + \sin xy \frac{\partial}{\partial y}(x) = x \cos xy \frac{\partial}{\partial y}(xy) + 0 = x^2 \cos xy$.

5. Given $f(x, y) = \frac{2y}{y + \cos x}$. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution: $\frac{\partial f}{\partial x} = \frac{(y + \cos x) \frac{\partial}{\partial x}(2y) - 2y \frac{\partial}{\partial x}(y + \cos x)}{(y + \cos x)^2} = \frac{2y \sin x}{(y + \cos x)^2}$.

$\frac{\partial f}{\partial y} = \frac{(y + \cos x) \frac{\partial}{\partial y}(2y) - 2y \frac{\partial}{\partial y}(y + \cos x)}{(y + \cos x)^2} = \frac{2 \cos x}{(y + \cos x)^2}$.

6. Given $z = f(x, y) = y \sin x + x \cos y$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Solution: $\frac{\partial z}{\partial x} = y \cos x + \cos y$ and $\frac{\partial z}{\partial y} = \sin x - x \sin y$.

7. Given $z = e^{xy}(2x - y)$. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

Solution: $\frac{\partial z}{\partial x} = e^{xy} \frac{\partial}{\partial x}(2x - y) + (2x - y) \frac{\partial}{\partial x}(e^{xy}) = 2e^{xy} + (2x - y)e^{xy} \frac{\partial}{\partial x}(xy)$
 $= [2 + y(2x - y)]e^{xy}$

Similarly, we can find $\frac{\partial z}{\partial y} = [-1 + x(2x - y)]e^{xy}$.

8. Given $u(x, y, z) = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$. Find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}$.

Solution: We have $u(x, y, z) = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$. Then

$$\frac{\partial u}{\partial x} = \frac{1}{y} + 0 - \frac{z}{x^2} = \frac{1}{y} - \frac{z}{x^2}, \quad \frac{\partial u}{\partial y} = -\frac{x}{y^2} + \frac{1}{z} \text{ and } \frac{\partial u}{\partial z} = -\frac{y}{z^2} + \frac{1}{x}.$$

$$\text{Hence } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = x\left(\frac{1}{y} - \frac{z}{x^2}\right) + y\left(-\frac{x}{y^2} + \frac{1}{z}\right) + z\left(-\frac{y}{z^2} + \frac{1}{x}\right) = 0.$$

9. Compute $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ for the functions (i) $u = e^{2x} \sin 3y$; (ii) $u = \log(x^3 + y^3 - 3xy)$.

Solution: (i) Given $u = e^{2x} \sin 3y$. So that $\frac{\partial u}{\partial x} = \sin 3y \frac{\partial}{\partial x}(e^{2x}) = 2e^{2x} \sin 3y$, and

$$\frac{\partial u}{\partial y} = e^{2x} \frac{\partial}{\partial y}(\sin 3y) = 3e^{2x} \cos 3y.$$

(ii) Given $u = \log(x^3 + y^3 - 3xy)$.

$$\text{So that } \frac{\partial u}{\partial x} = \frac{1}{x^3 + y^3 - 3xy} \frac{\partial}{\partial x}(x^3 + y^3 - 3xy) = \frac{3x^2 + 0 - 3y}{x^3 + y^3 - 3xy} = \frac{3(x^2 - y)}{x^3 + y^3 - 3xy}$$

$$\text{and } \frac{\partial u}{\partial y} = \frac{1}{x^3 + y^3 - 3xy} \frac{\partial}{\partial y}(x^3 + y^3 - 3xy) = \frac{0 + 3y^2 - 3x}{x^3 + y^3 - 3xy} = \frac{3(y^2 - x)}{x^3 + y^3 - 3xy}.$$

Exercises

Find $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ for the following functions:

(i) $u = 7x^3 - 5x^2y + 6y^3$;

(ii) $u = (2x - y)^5$;

(iii) $u = 3e^{-2x} \cos 2y + x^3 - 3xy^2 + 5z - 7$;

(iv) $u = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$;

(v) $u = x\sqrt{x + y}$.

We know that if $y = f(x)$ then $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$. Sometimes we write $\frac{dy}{dx}$ or D_y for $f'(x)$. Now if $z = f(x, y)$ then

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

and
$$\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.$$

In this case, we write $\frac{\partial z}{\partial x}$ or $D_x z$ for $\frac{\partial f}{\partial x}$ and $\frac{\partial z}{\partial y}$ or $D_y z$ for $\frac{\partial f}{\partial y}$.

Worked out problems

If $u(x, y) = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, then find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.

Solution: Given $u(x, y) = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$

$$\begin{aligned} \text{Then } \frac{\partial u}{\partial x} &= \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \frac{\partial}{\partial x} \left(\frac{x}{y} \right) + \frac{1}{1 + \frac{y^2}{x^2}} \frac{\partial}{\partial x} \left(\frac{y}{x} \right) = \frac{y}{\sqrt{y^2 - x^2}} \times \frac{1}{y} + \frac{x^2}{x^2 + y^2} \left(-\frac{y}{x^2} \right) \\ &= \frac{1}{\sqrt{y^2 - x^2}} - \frac{y}{x^2 + y^2} \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{1}{\sqrt{1 - \frac{x^2}{y^2}}} \frac{\partial}{\partial y} \left(\frac{x}{y} \right) + \frac{1}{1 + \frac{y^2}{x^2}} \frac{\partial}{\partial y} \left(\frac{y}{x} \right) = \frac{y}{\sqrt{y^2 - x^2}} \left(-\frac{x}{y^2} \right) + \frac{x^2}{x^2 + y^2} \left(\frac{1}{x} \right) \\ &= -\frac{x}{y\sqrt{y^2 - x^2}} + \frac{x}{x^2 + y^2} \end{aligned}$$

$$\text{Therefore, } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{x}{\sqrt{y^2 - x^2}} - \frac{xy}{x^2 + y^2} - \frac{xy}{y\sqrt{y^2 - x^2}} + \frac{xy}{x^2 + y^2} = 0.$$

Exercises

1. Given $v(x, y) = ax^2 + 2hxy + by^2$. Show that

$$\left(\frac{\partial v}{\partial x} \right)^2 \frac{\partial^2 v}{\partial y^2} - 2 \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} \frac{\partial^2 v}{\partial x \partial y} + \left(\frac{\partial v}{\partial y} \right)^2 \frac{\partial^2 v}{\partial x^2} = 8(ab - h^2)v(x, y).$$

2. If $v(x, y, z) = (x^2 + y^2 + z^2)^{\frac{m}{2}}$, then find the value of $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}$.

Chain rule for partial differentiation

(i) If $u = f(x, y, z)$ and $x = x(t), y = y(t), z = z(t)$ then

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} + \frac{\partial u}{\partial z} \frac{dz}{dt}$$

(ii) If $w = f(u, v)$ and $u = g(x, y, z), v = h(x, y, z)$ then

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial x}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial y}$$

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial z} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial z}$$

Worked out problems

If $w = f(u, v)$ and $u(x, y, z) = \frac{y-x}{xy}$, $v(x, y, z) = \frac{z-y}{yz}$, then prove that

$$x^2 \frac{\partial w}{\partial x} + y^2 \frac{\partial w}{\partial y} + z^2 \frac{\partial w}{\partial z} = 0.$$

Solution: Given $u(x, y, z) = \frac{y-x}{xy} = \frac{1}{x} - \frac{1}{y}$, $v(x, y, z) = \frac{z-y}{yz} = \frac{1}{y} - \frac{1}{z}$. Then

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial x} = -\frac{1}{x^2} \frac{\partial w}{\partial u} + 0 \times \frac{\partial w}{\partial v} \Rightarrow x^2 \frac{\partial w}{\partial x} = -\frac{\partial w}{\partial u}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial y} = \frac{1}{y^2} \frac{\partial w}{\partial u} - \frac{1}{y^2} \frac{\partial w}{\partial v} \Rightarrow y^2 \frac{\partial w}{\partial y} = \frac{\partial w}{\partial u} - \frac{\partial w}{\partial v}$$

$$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial u} \frac{\partial u}{\partial z} + \frac{\partial w}{\partial v} \frac{\partial v}{\partial z} = 0 \times \frac{\partial w}{\partial u} + \frac{1}{z^2} \frac{\partial w}{\partial v} \Rightarrow z^2 \frac{\partial w}{\partial z} = \frac{\partial w}{\partial v}$$

Thus, we get $x^2 \frac{\partial w}{\partial x} + y^2 \frac{\partial w}{\partial y} + z^2 \frac{\partial w}{\partial z} = 0$.

A function of the form

$$f(x, y) = a_n x^n + a_{n-1} x^{n-1} y + a_{n-2} x^{n-2} y^2 + \cdots + a_0 y^n$$

in which the sum of the powers of x and y in every term is the same (say, n) is called a *homogeneous function* in x, y of degree n .

Euler's theorem for homogeneous functions

If $f(x, y, z)$ is a homogeneous function in x, y and z of degree n having continuous first order partial derivatives then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = n f(x, y, z)$.

Worked out problems

1. If $u(x, y) = \tan^{-1} \frac{x^3 + y^3}{x - y}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$.

Solution: Given $u(x, y) = \tan^{-1} \frac{x^3 + y^3}{x - y}$. Then

$$\tan u = \frac{x^3 + y^3}{x - y} = \frac{x^3 \left\{ 1 + \left(\frac{y}{x} \right)^3 \right\}}{x \left\{ 1 - \left(\frac{y}{x} \right) \right\}} = x^2 \phi \left(\frac{y}{x} \right) \text{ where } \phi \left(\frac{y}{x} \right) = \frac{1 + \left(\frac{y}{x} \right)^3}{1 - \left(\frac{y}{x} \right)}.$$

Therefore, $\tan u$ is a homogeneous function in u of degree 2. Let $v = \tan u$.

Hence by Euler's theorem, we can write

$$\begin{aligned} x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} &= 2v \\ \Rightarrow x \left(\frac{\partial v}{\partial u} \frac{\partial u}{\partial x} \right) + y \left(\frac{\partial v}{\partial u} \frac{\partial u}{\partial y} \right) &= 2 \tan u \\ \Rightarrow x \sec^2 u \frac{\partial u}{\partial x} + y \sec^2 u \frac{\partial u}{\partial y} &= 2 \tan u \\ \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= 2 \frac{\sin u}{\cos u} \cos^2 u = 2 \sin u \cos u = \sin 2u. \end{aligned}$$

2. If $u = \tan^{-1} \frac{5x^4 - 3xy^3 + 7y^2z^2}{x^2 - 2y^2 + 3yz}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \sin 2u$.

Solution: Given $u = \tan^{-1} \frac{5x^4 - 3xy^3 + 7y^2z^2}{x^2 - 2y^2 + 3yz}$. Then

$$\tan u = \frac{5x^4 - 3xy^3 + 7y^2z^2}{x^2 - 2y^2 + 3yz} = \frac{x^4 \left\{ 5 - 3 \left(\frac{y}{x} \right)^3 + 7 \left(\frac{yz}{x^2} \right)^2 \right\}}{x^2 \left\{ 1 - 2 \left(\frac{y}{x} \right)^2 + 3 \left(\frac{yz}{x^2} \right) \right\}} = x^2 \phi \left(\frac{y}{x}, \frac{yz}{x^2} \right)$$

$$\text{where } \phi \left(\frac{y}{x}, \frac{yz}{x^2} \right) = \frac{5 - 3 \left(\frac{y}{x} \right)^3 + 7 \left(\frac{yz}{x^2} \right)^2}{1 - 2 \left(\frac{y}{x} \right)^2 + 3 \left(\frac{yz}{x^2} \right)}.$$

Therefore, $\tan u$ is a homogeneous function in u of degree 2. Let $v = \tan u$.

Hence by Euler's theorem, we can write

$$\begin{aligned}
& x \frac{\partial v}{\partial x} + y \frac{\partial v}{\partial y} + z \frac{\partial v}{\partial z} = 2v \\
\Rightarrow & x \left(\frac{\partial v}{\partial u} \frac{\partial u}{\partial x} \right) + y \left(\frac{\partial v}{\partial u} \frac{\partial u}{\partial y} \right) + z \left(\frac{\partial v}{\partial u} \frac{\partial u}{\partial z} \right) = 2 \tan u \\
\Rightarrow & x \sec^2 u \frac{\partial u}{\partial x} + y \sec^2 u \frac{\partial u}{\partial y} + z \sec^2 u \frac{\partial u}{\partial z} = 2 \tan u \\
\Rightarrow & x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 2 \sin u \cos u = \sin 2u.
\end{aligned}$$

Exercises

If $u = \tan^{-1} \frac{x^2 + y^2 + z^2}{x + y + z}$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{1}{2} \sin 2u$.

Laplace's equation

Laplace's equation is a second-order partial differential equation named after French mathematician and astronomer Pierre-Simon Laplace (from 23 March 1749 to 5 March 1827) who first studied its properties. This is often written as $\nabla^2 \phi = 0$ where ∇^2 is the Laplacean operator and ϕ is a scalar function.

The general theory of solutions to Laplace's equation is known as potential theory. The solutions of Laplace's equation are the harmonic functions which are important in many fields of science, notably the fields of electromagnetism, astronomy and fluid dynamics, because they can be used to accurately describe the behavior of electric, gravitational and fluid potentials. In the study of heat conduction, the Laplace's equation is the steady-state heat equation.

Worked out problems

1. Prove that the function $u(x, y) = x^2 - y^2 - 2xy - 2x + 3y$ satisfies the Laplace's equation.

Solution. We have $u(x, y) = x^2 - y^2 - 2xy - 2x + 3y$

$$\text{So } \frac{\partial u}{\partial x} = 2x - 2y - 2 \text{ and } \frac{\partial^2 u}{\partial x^2} = 2$$

$$\text{Again, } \frac{\partial u}{\partial y} = -2y - 2x + 3 \text{ and } \frac{\partial^2 u}{\partial y^2} = -2$$

$$\text{Thus } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \text{ i.e., } u(x, y) \text{ satisfies the Laplace's equation.}$$

2. Prove that the function $u(x, y) = e^{-x}(x \sin y - y \cos y)$ satisfies the Laplace's equation.

Solution. We have $u(x, y) = e^{-x}(x \sin y - y \cos y)$ so that

$$\frac{\partial u}{\partial x} = e^{-x}(\sin y) - e^{-x}(x \sin y - y \cos y) = e^{-x}(\sin y - x \sin y + y \cos y),$$

$$\frac{\partial^2 u}{\partial x^2} = e^{-x}(-\sin y) - e^{-x}(\sin y - x \sin y + y \cos y) = e^{-x}(-2 \sin y + x \sin y - y \cos y) \quad (1)$$

Again, $\frac{\partial u}{\partial y} = e^{-x}(x \cos y - \cos y + y \sin y) = e^{-x}(x \cos y - \cos y + y \sin y)$,

$$\frac{\partial^2 u}{\partial y^2} = e^{-x}(-x \sin y + \sin y + y \cos y + \sin y) = e^{-x}(2 \sin y - x \sin y + y \cos y) \quad (2)$$

Adding (1) and (2), $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, i.e., $u(x, y)$ satisfies the Laplace equation.

3. If $v = f(x, y)$, where $x = r \cos \theta$, $y = r \sin \theta$, prove that

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2}.$$

Solution: Given $x = r \cos \theta$, $y = r \sin \theta$ where $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$.

We have $v = f(x, y)$. Therefore,

$$\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2 + y^2}} = \frac{x}{r} = \frac{r \cos \theta}{r} = \cos \theta,$$

$$\frac{\partial r}{\partial y} = \frac{2y}{2\sqrt{x^2 + y^2}} = \frac{y}{r} = \frac{r \sin \theta}{r} = \sin \theta,$$

$$\frac{\partial \theta}{\partial x} = \frac{-\frac{y}{x^2}}{1 + \frac{y^2}{x^2}} = -\frac{y}{x^2 + y^2} = -\frac{r \sin \theta}{r^2} = -\frac{\sin \theta}{r},$$

$$\frac{\partial \theta}{\partial y} = \frac{\frac{1}{x^2}}{1 + \frac{y^2}{x^2}} = \frac{x}{x^2 + y^2} = \frac{\cos \theta}{r}.$$

$$\text{Hence, } \frac{\partial v}{\partial x} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial v}{\partial \theta} \frac{\partial \theta}{\partial x} = \cos \theta \frac{\partial v}{\partial r} - \frac{\sin \theta}{r} \frac{\partial v}{\partial \theta} \quad (2)$$

$$\text{and } \frac{\partial v}{\partial y} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial v}{\partial \theta} \frac{\partial \theta}{\partial y} = \sin \theta \frac{\partial v}{\partial r} + \frac{\cos \theta}{r} \frac{\partial v}{\partial \theta} \quad (3)$$

$$\text{We can write } \frac{\partial}{\partial x} \equiv \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \quad (4)$$

$$\text{and } \frac{\partial}{\partial y} \equiv \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \quad (5)$$

$$\begin{aligned} \text{So } \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial x} \right) &= \frac{\partial^2 v}{\partial x^2} = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos \theta \frac{\partial v}{\partial r} - \frac{\sin \theta}{r} \frac{\partial v}{\partial \theta} \right) \\ &= \cos \theta \left[\cos \theta \frac{\partial^2 v}{\partial r^2} - \frac{\sin \theta}{r} \frac{\partial^2 v}{\partial r \partial \theta} + \frac{\sin \theta}{r^2} \frac{\partial v}{\partial \theta} \right] \end{aligned}$$

$$\begin{aligned}
& -\frac{\sin \theta}{r} \left[\cos \theta \frac{\partial^2 v}{\partial \theta \partial r} - \sin \theta \frac{\partial v}{\partial r} - \frac{\cos \theta}{r} \frac{\partial v}{\partial \theta} - \frac{\sin \theta}{r} \frac{\partial^2 v}{\partial \theta^2} \right] \\
& = \cos^2 \theta \frac{\partial^2 v}{\partial r^2} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 v}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 v}{\partial \theta^2} + \frac{\sin^2 \theta}{r} \frac{\partial v}{\partial r} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial v}{\partial \theta}
\end{aligned} \quad (6)$$

Similarly, we can show that

$$\frac{\partial^2 v}{\partial y^2} = \sin^2 \theta \frac{\partial^2 v}{\partial r^2} + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 v}{\partial r \partial \theta} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 v}{\partial \theta^2} + \frac{\cos^2 \theta}{r} \frac{\partial v}{\partial r} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial v}{\partial \theta} \quad (7)$$

$$\text{Thus} \quad \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} = 0 \quad (8)$$

which is the Laplace's equation in polar coordinates. Solution of Laplace's equation in polar coordinates is called *circular harmonics*.

Exercises

1. Show that the following functions satisfy the Laplace's equation:

- (i) $u(x, y) = y^3 - 3x^2 y$
- (ii) $u(x, y) = 3x^2 y + 2x^2 - y^3 - 2y^2$
- (iii) $u(x, y) = e^x (x \cos y - y \sin y)$
- (iv) $u(x, y) = \frac{1}{2} \log(x^2 + y^2)$
- (v) $u(x, y, z) = (x^2 + y^2 + z^2)^{-\frac{1}{2}}$
- (vi) $u(x, y, z) = 3e^{-2x} \cos 2y + x^3 - 3xy^2 + 5z - 7.$

2. If $v = f(x, y)$ with $x = r \cos \theta$ and $y = r \sin \theta$, then express $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$ in polar coordinates.
